

Concept Review

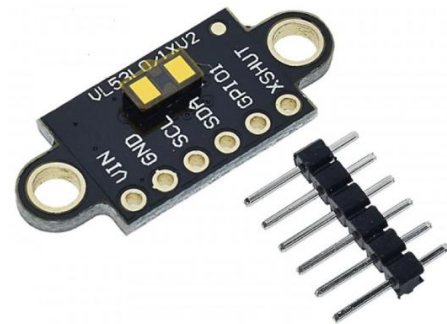
Time of Flight

Why Do We Need Time of Flight Sensors?

Measuring the distance between any two points can be estimated using various techniques. Time of Flight sensors fall under the topology of distance measuring devices. Like the name suggests, Time of Flight sensors use the property of sending out a pulse and measuring how long it takes to return to a sensor at which point a distance measurement can be estimated.

Time of Flight

Time of Flight sensors utilize the concept of signal emission and signal detection. By estimating the amount of time taken for the emitted signal to be detected back at the sensor a distance measurement can be estimated.



a. Tiny Electronics VL53L1X



b. Adafruit TFmini

Figure 1: Different configurations for laser-based Time of Flight sensors.

Figure 2 shows an example of a laser-based time of flight sensor which estimates the straight-line distance to a subject of interest.

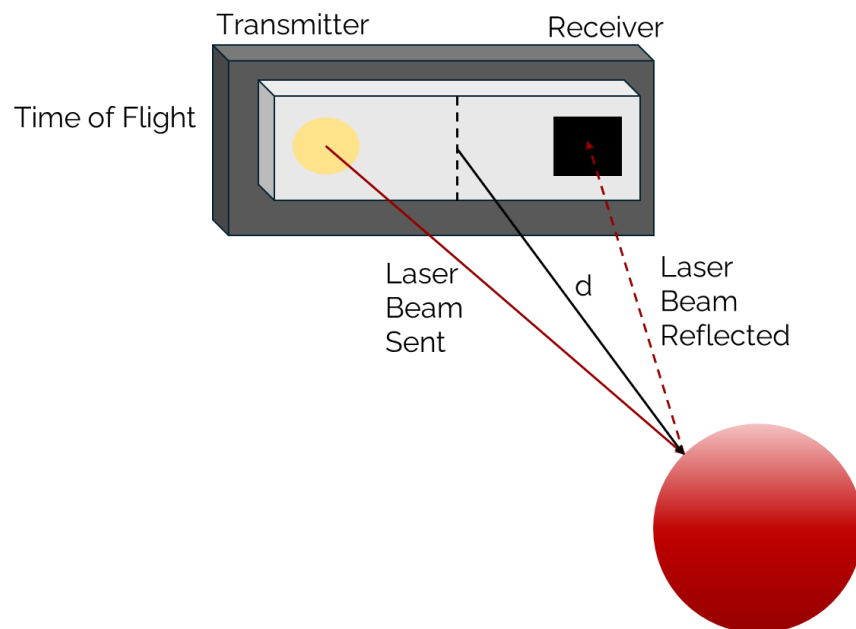


Figure 2: Distance estimate from a laser-based Time of Flight sensor

At the fundamental level a laser emitting time of flight sensor looks at the time taken for light to hit an object and reflect to the receiver. Knowing the speed of light and the time between the sensor and the emitter we can use the following function to estimate the distance to an object:

$$d = c\Delta t/2 \quad (1)$$

In equation 1, the constant c is 299,792,458 m/s. Depending on the sensor manufacturer, this calculation may be done within the sensor directly. As an example, the STM VL53L8CHV0GC will provide distance values directly.

Another interesting characteristic of modern Time-of-Flight sensors is multizone measurements. Multizone refers to a grid like series of measurements which provide more information than a single beam. This information can be useful if you're interested in interpreting the shape of an object within proximity to the Time-of-Flight sensor. Sensors such as the STM VL53L8CHV0GC can provide you with multizone distance values.

Time of Flight sensors also have different technical characteristics based on the manufacturer's specifications. Examples of different Time of Flight sensor technologies:

- VL53L8CHV0GC uses a single photon avalanche diode (SPAD) detection array.
- ADSD3100 uses a Complementary Metal Oxide Semiconductor (CMOS) sensor.

For a VL53L8CHV0GC the SPAD detector array can generate a multizone distance measurement as seen in Figure 3. For this example, a maximum distance of 0.5m is represented by the dark red and a bright blue are values close to the sensor array.

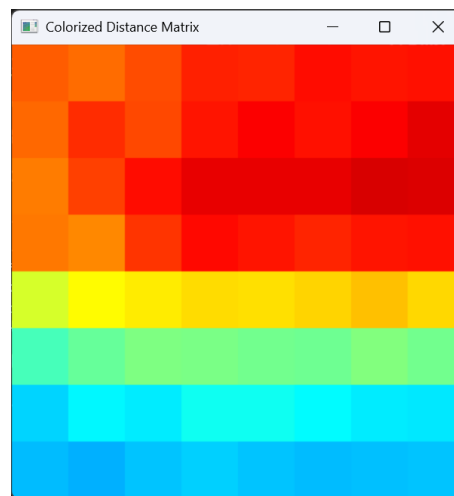


Figure 3: Colorized multizone distance information from VL53L8CHV0GC

© Quanser Inc., All rights reserved.



Solutions for teaching and research. Made in Canada.